ANUSHA ANAND

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SUMMARY

Sr. Software Engineer specializing in data engineering for financial systems. Built PySpark and Kafka pipelines processing 10M+ records daily at Citi, migrated on-premises data jobs to AWS using Airflow and Iceberg, and cut data-quality failure detection from hours to under 10 minutes. Earlier background in Java and Spring banking platforms at Nomura. AWS certified in Solutions Architecture and Machine Learning.

SKILLS

Data and Cloud: Spark, PySpark, Kafka, Airflow, Iceberg, Databricks, AWS (S3, Lambda, API Gateway, CloudWatch), Docker, Jenkins, CI/CD, Git
Languages: Python, SQL, Java, JavaScript
Backend and Databases: FastAPI, Flask, Spring Boot, Node.js, REST APIs, PostgreSQL, MySQL
AI and Machine Learning: Claude API, OpenAI API, LangChain, RAG, Embeddings, FAISS, Vector Search

PROFESSIONAL EXPERIENCE

Sr. Software Engineer, Client Engagement - Citigroup | Tech Mahindra Americas Inc. **Feb 2025 – Present**

- Architected Python and PySpark ETL pipelines using Apache Spark and Kafka processing 10M+ financial records daily, reducing manual validation effort by 35%.
- Migrated on-premises data jobs to AWS S3, orchestrating with Apache Airflow and managing tables with Apache Iceberg.
- Built an AI-driven data-quality framework combining anomaly detection and rule-based flagging across 5+ workflows, reducing defect escapes by 28%.
- Implemented CloudWatch dashboards and alerts to monitor, diagnose, and resolve data-quality issues, cutting failure detection from hours to under 10 minutes.
- Developed monitoring/validation pipelines for wealth management that translate business requirements into technical solutions, track money movements, and flag unusual activity.
- Engineered an ML test-generation system that mines historical failure logs to produce targeted test cases, increasing QA coverage by 40%.
- Delivered 3+ concurrent projects from planning through production with no bug escalations, earning engineering excellence recognition and a Star Award nomination.

Software Engineer Intern | iHeartMedia Inc. **Jun 2024 – Aug 2024**

- Designed and deployed an AWS CDK-based serverless scheduling solution using Lambda, API Gateway, S3, SNS, and SQS, automating 20+ integration workflows and reducing manual operations by 90%.
- Secured 8+ API Gateway endpoints with custom Lambda authorizers and JWT-based access control, eliminating unauthenticated access across production services.
- Built React.js dashboards featuring filtering and six real-time metrics across 5,000+ records, replacing manual reporting processes.

Associate Software Engineer II | Nomura Research Institute Financial Technologies **Apr 2022 – Jul 2022**

- Developed four core banking modules using Java, Spring, and Oracle SQL to support 500K+ daily retail and corporate transactions in a high-reliability environment.
- Refactored Java batch processing and ETL workflows, reducing pipeline latency by 25% and cutting nightly runtime from 4 hours to under 3.
- Selected for a three-month onsite assignment in Japan to lead client-side development and align deliverables with stakeholder expectations.

Associate Software Engineer I | Nomura Research Institute Financial Technologies **Aug 2020 – Mar 2022**

- Led the design and development of 6+ enterprise banking and asset management applications using Java 8/11, Spring MVC, and SQL within Agile sprints.
- Integrated unit, integration, and end-to-end testing into release workflows, cutting production defects by 30% and release cycles by two days.
- Mentored 5+ junior developers in scripting, batch development, and Agile practices.

PROJECTS

Real-Time CDC Lakehouse for Payments Analytics *(Debezium, Kafka, Spark, Delta Lake, dbt)* **Oct 2026**

- Built a real-time CDC pipeline streaming PostgreSQL inserts, updates, and deletes through Debezium and Kafka into Delta Lake Bronze (change log) and Silver (current state) layers using Spark Structured Streaming and idempotent MERGE upserts.
- Modeled a dbt star schema with SCD Type 2 customers and point-in-time joins; added 20 automated tests, a quarantine table, and CI.

YouTube Video Q&A Assistant *(Streamlit, LangChain, FAISS, OpenAI)* **Feb 2025 – May 2025**

- Built a retrieval-augmented question answering application that embeds YouTube transcripts into FAISS and answers questions with context-grounded responses from OpenAI GPT.

EDUCATION

Master of Science in Computer Science, The University of Texas at Arlington (GPA 4.0/4.0) **Aug 2022 – May 2024**
Bachelor of Technology in Computer Science, Sikkim Manipal Institute of Technology (GPA 9.2/10) **Aug 2016 – Jul 2020**

CERTIFICATIONS

Databricks Certified Data Engineer – Associate **In Progress**
AWS Certified Machine Learning Engineer – Associate **Oct 2025 – Oct 2028**
AWS Certified Solutions Architect – Associate **Dec 2024 – Dec 2027**